

NAWABSHAH, Pakistan

WMO: 417490

Lat: 26.219N

Lon: 68.390E

Elev: 28

StdP: 100.99

Time Zone: 5.00 (E05)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	7.0	1.0	4.1	14.9	2.2	4.5	16.7	6.0	18.8	5.3	19.9	0.9	0	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.9	46.8	25.5	45.1	25.5	43.8	25.6	29.5	39.0	29.0	38.8	28.5	38.4	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	34.4	26.7	22.4	34.3	26.1	21.5	34.1	97.9	39.2	95.0	39.2	92.8	38.5	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.4	5.7	DB	3.6	49.0	2.4	0.8	1.8	49.6	0.4	50.1	-1.0	50.6	-2.8	51.2	(4)
(5)			WB	2.5	30.7	2.3	0.7	0.8	31.2	-0.5	31.6	-1.9	32.0	-3.6	32.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	15.7	19.4	25.5	31.3	35.4	35.8	34.5	32.9	32.2	29.2	22.9	17.4	(6)	
	DBStd	7.31	2.01	3.00	3.21	2.51	1.89	1.64	1.62	1.44	1.24	2.18	2.62	2.37	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	154	84	21	1	0	0	0	0	0	0	0	1	47	(9)	
	CDD10.0	6466	177	263	480	639	788	775	759	709	667	594	388	228	(10)	
	CDD18.3	3579	3	50	223	389	529	525	501	451	417	336	139	17	(11)	
	CDH23.3	54454	146	736	3002	5967	8813	8685	7918	6722	6047	4448	1652	319	(12)	
(13)	CDH26.7	36018	15	263	1803	4101	6421	6297	5446	4266	3781	2782	785	58	(13)	
(14)	Wind	WSAvg	2.6	1.7	1.9	2.1	2.5	3.1	4.2	3.7	3.2	2.0	1.4	1.5	(14)	
(15) Precipitation	PrecAvg	118	1	2	3	2	1	10	37	38	10	0	2	4	(15)	
	PrecMax	415	10	8	32	39	9	109	199	135	69	6	25	29	(16)	
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	109	2	3	7	8	2	25	43	47	19	1	5	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	34.2	42.1	46.2	48.9	48.1	44.1	41.8	42.0	41.2	36.9	30.2	(19)	
		MCWB	17.2	19.8	22.1	23.9	25.4	26.5	28.2	27.9	26.4	24.2	21.5	18.7	(20)	
	2%	DB	26.2	32.1	39.9	44.2	47.1	46.0	42.8	40.2	40.6	40.0	34.9	28.2	(21)	
		MCWB	16.1	19.5	21.7	23.5	25.3	26.3	28.0	27.9	26.7	24.3	21.4	17.6	(22)	
	5%	DB	25.0	30.0	37.5	42.8	45.9	44.2	41.5	39.4	39.2	38.9	33.1	26.8	(23)	
		MCWB	15.6	18.4	21.2	23.3	25.2	26.9	28.0	27.7	26.6	23.9	20.7	16.8	(24)	
	10%	DB	23.3	27.9	35.1	41.0	44.2	42.9	40.3	38.2	38.2	37.2	31.2	25.0	(25)	
		MCWB	14.6	17.3	20.7	22.9	25.2	27.0	28.0	27.6	26.5	23.6	19.9	15.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	21.8	24.9	27.2	29.1	29.7	30.3	29.7	29.3	27.4	23.8	19.9	(27)	
		MCDB	26.3	30.9	36.2	41.9	42.9	40.6	39.3	37.9	38.1	37.2	32.6	26.2	(28)	
	2%	WB	17.1	20.3	23.3	25.8	28.1	29.1	29.6	29.1	28.5	26.4	22.6	18.6	(29)	
		MCDB	25.1	30.0	35.6	40.2	42.1	40.3	38.9	37.3	37.2	36.1	31.9	26.4	(30)	
	5%	WB	16.1	19.1	22.2	24.7	27.4	28.6	29.0	28.6	27.8	25.6	21.6	17.6	(31)	
		MCDB	23.7	28.5	34.8	39.0	41.0	40.0	38.8	36.9	36.3	35.2	30.5	25.3	(32)	
	10%	WB	15.2	18.0	21.2	23.8	26.7	28.1	28.5	28.1	27.2	24.9	20.6	16.6	(33)	
		MCDB	22.3	26.6	33.4	38.0	39.9	39.7	38.2	36.4	35.8	34.3	29.4	23.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.9	18.3	19.0	18.4	14.9	12.2	11.4	13.3	17.0	16.9	15.9	(35)	
		MCDBR	18.1	19.3	21.5	21.5	20.2	17.3	13.8	12.9	14.6	18.3	18.4	17.3	(36)	
	5% WB	MCWBR	9.2	8.5	7.4	6.5	5.4	4.8	4.4	4.5	4.9	5.7	7.3	8.1	(37)	
		MCWBR	17.0	17.6	19.6	19.9	18.0	14.4	12.2	11.5	12.7	15.7	16.9	16.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.472	0.480	0.539	0.607	0.649	0.717	0.784	0.717	0.612	0.562	0.565	0.527	(40)	
		taud	1.914	1.933	1.805	1.698	1.654	1.570	1.504	1.606	1.761	1.772	1.726	1.786	(41)	
	Ebn at Noon		753	789	766	726	696	646	603	641	698	703	656	673	(42)	
		Edh at Noon	169	178	214	244	254	275	294	264	220	207	203	186	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.02	4.91	5.95	6.66	7.07	6.76	6.00	5.91	5.87	5.34	4.25	3.81	(44)	
	RadStd		0.19	0.22	0.22	0.18	0.11	0.19	0.36	0.27	0.29	0.22	0.19	0.18	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page